

Usefulness of Galactose Oxidase-Schiff Test in Rectal Mucus for Screening of Colorectal Malignancy*

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Abstract. Based on a "field-effect" theory in colon carcinogenesis, and the expression of the disaccharide tumor marker D galactose- β -[1 $\rightarrow$ 3]-N-acetyl-D-galactosamine (Gal-GalNAc) in the rectal mucus of patients with cancer and precancer of the colon, Shamsuddin developed a simple, accurate, inexpensive, easy to perform and rapid (≤ 15 min) screening test for colonic cancer and precancerous lesions. In this study we examined 137 rectal mucus samples of randomly selected patients with colorectal malignancy or other colorectal diseases to confirm the sensitivity and specificity of this test in Croatia. Additionally, to test the validity of the "field-effect" theory, that the mucosa away from the obvious cancer will show abnormalities as a result of the generalized effect of the carcinogen throughout the entire field of the target tissue, we also monitored a subset of 53 patients post-operatively. Individuals free of colonic or any other malignancies served as control ($n=31$). The rectal mucin was smeared on membrane filter and developed by a sequential reaction of galactose oxidase (GO) and Schiff's reagent. The test results were correlated to the findings from colonoscopy/surgery and histopathology. The sensitivity of the test was shown to be 100% and the specificity was 96.8% ($p<0.001$). Interestingly, the test was positive in 60% (32 of 53) of the samples collected from patients after tumor resection, showing the persistence of the biochemical changes even though malignant tumors were removed, hence supporting the field-effect phenomenon of carcinogenic stimuli. Five patients out of these 32 (16%) post-operative cases with positive GO test had a tumor recurrence within a year ($0.05<p<0.10$), suggesting that a persistently positive GO test in this population may serve as a predictor of tumor recurrence. We conclude that Gal-GalNAc is an early and intermediate biomarker, suitable not only for the detection of

malignancy in its inception, but also for monitoring of people at high risk for cancer, and the efficacy of the cancer therapy as well as secondary prevention by this technology.

Early detection and prevention are still our best strategies to fight various cancers, including that of the colon. Unfortunately, this effort has been hampered because of the insensitivity and non-specificity of the existing screening assays, i.e. the fecal occult blood tests (FOBT). This is not surprising, since blood in stool, occult or otherwise is not a reliable marker of malignancy. Mucins are high molecular weight and heavily glycosylated glycoproteins. They are produced by epithelial cells of normal and malignant tissues, either as secreted or membrane-bound molecules [1,2]. Development and progression of cancer are almost always accompanied by changes in the biochemical characteristics of mucins, including both abnormal glycosylation of mucin core peptides and an altered expression of mucin genes [1-4].

The carbohydrate moiety D-galactose- β -[1 $\rightarrow$ 3]-N-acetyl-D-galactosamine (Gal-GalNAc), is an immunodeterminant structure of the Thomsen-Friedenreich (TF) antigen, and also a precursor of the M and N blood group substances (although it may be different). Gal-GalNAc has been expressed by the mucin of colon cancer cells, by the precancerous lesions of the colon, as well as by the normal-appearing mucosa remote from carcinoma, but not by normal mucin; in normal colorectal mucosa this structure is masked by oligosaccharide side-chain extension and/or sialylation [5-10]. Being differentially expressed in normal and high-risk tissue and appearing during the very early stages of progression of carcinogenesis, Gal-GalNAc can serve as a reliable biomarker of precancer and cancer of the colon.

The altered expression of colonic mucin is observed not only in carcinomas, but also in morphologically normal-appearing mucosa distant from the carcinomas both in humans and in rodents treated with carcinogens. It is difficult to relate these changes to the genesis of cancer within the mindset of the prevalent dogma that most colorectal cancers arise from pre-existing polyps. This expression pattern

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supports the field-effect theory of carcinogenesis, suggested by Shamsuddin and based on his extensive studies in comparative pathology [10-12]. He hypothesized that the abnormalities in the morphologically normal-appearing mucosa are the result of the generalized effect of the carcinogen throughout the entire field of the target tissue, in this case colonic epithelium [10-12]. The existence of field-effect phenomenon in cancer had been first proposed by Slaughter *et al.* in the oral cavity [13]. It is believed that the carcinogenic agent affects a wide area of the target tissue, and initiates many cells, that are further subjected to a variable degree of promotion and progression, resulting in multifocal changes of initiated, but not fully promoted foci. Therefore, the expression of Gal-GalNAc may be one of the earliest phenotypic changes that precedes morphological changes to cancer, and therefore predictor of the cancer. The existence of field-effect phenomenon may not only be important for cancer screening, but also be relevant for monitoring patients who have undergone total resection of the lesion for recurrence of cancer.

Gal-GalNAc can be detected by lectin probes, monoclonal antibodies, or by simple enzymatic oxidation with galactose oxidase (GO) followed by Schiff's reagent (GO-Schiff) [6-10,14,15]. GO specifically oxidizes C-6 hydroxyl groups of D-galactopyranose and N-acetylgalactosamine residues of Gal-GalNAc, generating two vicinal aldehydes that react with basic fuchsin (Schiff's reagent) to give magenta/purple coloration (Figure 1, Panel A). However, it seems that the detection of Gal-GalNAc varies with the technique used; compared to the lectin jacalin staining and immunostaining with antibody to T-antigen, GO reaction was more sensitive and more specific [15].

Easy and simple enzymatic detection of Gal-GalNAc served as a basis for the development of GO-Schiff assay, a simple, non-invasive and rapid screening test for early detection of colorectal neoplasia and preneoplastic condition [16-18]. In summary, the test is based on the facts that (a) the altered mucin is not only expressed by cancerous and precancerous cells, but exists in morphologically normal epithelia and secretions in the vicinity of carcinoma, as well as in remote mucosa away from cancer, (b) this phenomenon may be the result of previous exposure of the host to the carcinogen (the concept of the "field-effect" theory), and (c) since mucin is secreted by the colorectal mucosa and can be easily sampled from the rectum, identification of such carbohydrate abnormalities in rectal mucus can be exploited for early detection of colorectal neoplasia [16-18].

Studies of this test in screening populations for colorectal cancer have shown that this tumor marker in the rectal mucus of patients with colorectal malignancies or pre-malignant conditions and lesions can be identified with high accuracy, suggesting its usefulness in mass screening. Evaluation of this test conducted in USA, Europe, Japan and in China on thousands of people have confirmed its high specificity and high sensitivity, despite racial, ethnic or other differences [19-34].

The purpose of this study was (a) to confirm the specificity and sensitivity of GO test in patients with colorectal cancer and precancerous lesions in Croatia, and (b) to monitor patients after total resection of colorectal malignancy to test for the existence of field-effect phenomenon and to get insight into recurrence of cancer.

Materials and Methods

Chemicals. Galactose oxidase type V (GO) and Schiff's reagent were obtained from Sigma Chemical, Co. (St. Louis, Missouri). Matriel nitrocellulose membrane filters were purchased from Gelman Sciences, Inc. (Ann Arbor, Michigan). Membrane filters were cut into approximately 1x1 cm squares for sample collection.

Patients. Samples of rectal mucus were obtained from the Gastroenterology Department in the Clinical Hospital, Split, Croatia. Altogether, 137 samples of patients with colorectal symptoms were investigated. The samples were collected from 65 colorectal carcinoma, 19 colorectal polyps, and 53 from the patients who had undergone total resection of the colorectal lesions previously. All patients with carcinoma and polyps underwent surgical procedures, followed by histological examination and categorization of the removed carcinoma and polyps. For clinico-pathological characteristics, carcinomas staged according to the system of Dukes. In this group, 21% of carcinomas were stage A, 29 % B, 21% C and 29% stage D. Most of the carcinomas were located in the rectum (45, 69%) and left colon (12, 19%), while only 5 (8%) were found in the right and 3 (5%) in transverse colon (Figure 2). Among the patients with polyps, 3 had familial adenomatous polyposis. None of the patients with polyps had hyperplastic polyps, all were identified as adenomas. The location of adenomas were similar as carcinomas; most were located in the rectum and left colon, 16 (84%) and 3 (16%) respectively (Figure 2). According to the histological diagnosis, 72% of the adenomas were classified as tubular, 14% as tubulovillous, and 14% as villous. Control group comprised 31 healthy asymptomatic subjects, among them 8 had chronic prostatitis.

Samples were collected by digital rectal examination with gloved index finger moistened with normal saline. The mucus on the examining finger was smeared on the scored side of a membrane, air dried at room temperature, and stored in dark and dry environment until developed. Dried samples can be stained either immediately or at a later time; they appear to be stable and do not lose reactivity for up to 5 years in storage.

Performance of the test (Staining). GO was reconstituted in PBS, pH 7.2, to obtain 100 U/mL; 50 μ L was placed on each dried mucus samples for 10 min at room temperature. Filters were washed with deionized distilled water for 20 sec and then reacted with Schiff's reagent (1% basic fuchsin) for 60 sec. After being washed in running tap water for a few minutes, the samples were air dried and evaluated for color reaction (Figure 1, Panel B).

Evaluation of staining. Stained samples were interpreted blindly by one of the authors (I.V.), who was unaware of the diagnoses and the treatment the patients received, as the clinical status of each patient was not made available until the results of the test were evaluated and scored. A positive reaction was indicated by magenta coloration of the mucus on membrane filter (Figure 1, Panel C). Stain intensity varied from no apparent stain (0), light pink (1+), very pink (2+), magenta (3+), to intense magenta color (4+). Along with each batch of experimental samples, positive and negative controls were concurrently processed for quality control. Imprints of mucinous adenocarcinomas of the colon that are known to be positive, were used as positive control (Figure 1, Panel Da). As additional control, negative samples were tested for the presence of mucin (Figure 1, Panel Ded). To determine

whether a negative result was due to the absence of Gal-GalNAc or due to the absence of mucus itself (technical-negative), negative samples were treated with periodic acid followed by Schiff reaction, as periodic acid will yield vicinal aldehydes irrespective of the hexose type, thereby confirming the presence of mucin (Figure 1, Panel Db).

Statistical analysis. The chi square (χ^2) test was used to compare the difference between the two groups, neoplastic versus non-neoplastic. The performance of a diagnostic test is defined by its sensitivity, specificity, positive and negative predictive values, and efficiency. Diagnostic sensitivity (ST) is the proportion of diseased subjects who yield a positive test results. Specificity (SP) is a measure of frequency of a negative test in the absence of particular disease. Both are expressed in percent and are calculated from $ST = TP / (TP + FN) \times 100$ and $SP = TN / (TN + FP) \times 100$, where TP = true positives, FN = false negatives, TN = true negatives, and FP = false positives. The predictive value of a positive test is the percentage of all positive results that are true positives, or the frequency of patients with positive test results that are diseased $TP / (TP + FP) \times 100$. The predictive value of a negative result is the percentage of all negative results that are true negatives, or the frequency of patients with negative test results that are nondiseased $TN / (TN + FN) \times 100$. The efficiency of the test defines the percentage of patients who are correctly classified by the test as diseased or nondiseased $TP + TN / (TP + FP + FN + TN) \times 100$ [35].

Results

Samples of rectal mucus were obtained from 137 patients and 31 normal controls during digital examination and were tested by GO-Schiff method.

Overall summary of results from all cases are given in Table I. All patients with colorectal carcinoma and polyps expressed GO-reactive carbohydrates in rectal mucus, in contrast to control group ($p < 0.001$), showing 100% correspondence between GO reactivity and the presence of colonic neoplasm or preneoplastic conditions. In this study, GO test was positive in 100% of diseased patients and negative in 96.8% of nondiseased patients. Therefore, the sensitivity of the GO test for colorectal carcinoma was 100%; the specificity of the test was 96.8%. Predictive value of positive test was 98.8%, predictive value of negative test was 100%, giving the 99.1% efficiency of the test. In addition to patients with cancer and polyp, we tested several patients with ulcerative colitis and Crohn's disease, both known as predisposing conditions for colorectal cancer. Interestingly, we found that all 5 patients with ulcerative colitis expressed GO reactivity (100%), while of the 4 patients with Crohn's disease, 3 expressed GO reactivity in rectal mucus (75%).

Examples of GO-stained membranes containing samples of rectal mucus from one group of randomly selected patients are shown in Figure 1, Panel C. The clinical status of each patient from this group and the scored results of GO reactivity are given in Table II. Colorectal carcinoma was found in 33 cases (24 carcinoma of rectum, 7 of sigmoid colon, 1 of descending colon, 1 of transverse colon), and one sarcoma of the rectum. Polyps were found in 7 cases, 4 with dysplasia. There were 3 cases of post-colorectal surgery, one patient with ulcerative colitis and one with subacute proctitis in this group. The difference in the test results between

Table I. Distribution of GO reactivity in rectal mucus of patients with colorectal cancer and polyp and control group free of colonic disease.

Patient category	GO Test Positive GO	Test Negative
Colorectal cancer and polyp (n=84)	84/84 (100%)	0 (0%)
Free of neoplastic disease (n=31)	1/31* (3%)	30/31 (96.8%)

* 1 case of subacute proctitis was positive. However, in the follow-up study a sigmoid polyp was found in this patient.

Difference between the two groups of patients is significant at $p < 0.001$ (χ^2 test).

neoplastic and non-neoplastic group was significant at $p < 0.001$. Filters #7 and #29-33 were negative and from patients with chronic prostatitis. Note differences in the intensity of magenta coloration, #5 (sigmoid polyp), #11 (carcinoma of the rectum) and #40 (carcinoma of the colon) being the most intensive (4+). Fecal contamination and possible interference were present at #24 and #52. Because of massive fecal contamination, sample #51 was excluded from evaluation. Patient #1 with ulcerative colitis has been seen for the past eight years by gastroenterologist. A positive reaction in this case could be expected, since ulcerative colitis, an inflammatory bowel disease, is a predisposing condition for colonic cancer. Therefore, it will be useful to continue testing this patient with GO test, together with follow-up biopsy. Case #2 is very interesting; at the time of testing the patient presented with subacute proctitis, indicating a false positive reaction. However, a sigmoid polyp was found during follow-up, explaining the reactivity of the test in the initial screening.

When samples collected from patients who have undergone surgery to remove colorectal malignancies were analyzed, 60% showed positive GO reactivity (Table III). The prognosis following surgery for colorectal carcinoma largely depends on the extent of tumor spread. The expression of Gal-GalNAc indicated the existence of an abnormal colorectal mucosa, and that these individuals were at high risk. Within one year, 5 patients out of 32 (16%) in the group with positive GO reactivity had tumor recurrences, while none from the group with negative GO test ($0.05 < p < 0.10$), suggesting the utility of the GO test in screening for recurrent colorectal cancer as well.

Discussion

Colorectal carcinoma is still a leading cause of morbidity and mortality in the United States and the industrialized Western countries [36]. Despite many improvements in chemotherapy and surgery, its prognosis has been only slightly affected.

Therefore, the World Health Organization, the American Cancer Society and the Agency for Health-Care Policy and Research recommend screening for colorectal cancer, aiming to detect the lesion in asymptomatic stage, during its infancy, in order to reduce the incidence and mortality from colorectal cancer. These recommendations augment the development of the colorectal cancer screening guidelines regarding the patient population, risk, frequency of screening, reimbursement, the quality and attractiveness of screening tests for patients and physicians, preferences and acceptance by patients [37-43].

To be classified as a good and practical, the screening test needs to be accurate, reliable, non-invasive, and cost-effective. The most commonly used ones, fecal occult blood tests (FOBT), barium enema-X rays, and endoscopic visualization, have many limitations and do not satisfy these criteria [37-45].

The major flaws for colonoscopy, flexible sigmoidoscopy, barium enema X-ray procedures, or novel computed tomography and magnetic resonance colonography are cost-effectiveness and safety [40-42]. These techniques are very costly, thereby posing tremendous burden to the national health care cost. Besides, they are associated with subject discomfort, and potential risk and complications, particularly if performed by nonphysicians in order to reduce the cost of procedure. Moreover, many recent articles suggest that small flat colorectal adenomas with high malignant potential are common and may be missed during conventional colonoscopy [46,47].

Although it may seem safe and simple, FOBT is highly inaccurate. With high rate of false-positive outcomes, FOBT exposes many subjects to the potential complications of colonoscopy, and the high rate of false-negative tests gives patients with colorectal cancer a false sense of security and delays their proper diagnostic work-up [44]. Although rectal all bleeding is associated with colorectal cancer, it does not occur in all cases and is usually a symptom of minor colorectal disorders. Only a very small number of cases of colorectal cancer present with rectal bleeding in isolation [45]. Actually, it was shown that in patients with positive FOBT, the upper gastrointestinal lesions were identified more frequently than colonic lesions [48]. Thus, since neither all cancers bleed nor all bleedings are due to cancer, the most important criticism of this method is the assumption that blood in the stool is being a marker for cancer, that it is not.

On the contrary, colorectal mucin can be utilized as a tumor marker. It has been well documented that alterations in mucin glycoconjugates occur during development of cancer. The disaccharide Gal-GalNAc is differentially expressed in high risk and normal tissue, and the presence of Gal-GalNAc in rectal mucin would imply the existence of an abnormal mucosa somewhere in the colorectum. The detection of this tumor marker in rectal mucus by enzymatic oxidation was the basis for development of a simple, rapid (≤ 15 min), inexpensive, and noninvasive test, with a great potential for mass screening [16-18]. The validity of GO test in screening the asymptomatic population was evaluated in

the United States, Japan and China on thousands of cases (more than 10,000) [19-34]. The results of the present study are consistent with previous reports, showing high efficiency and diagnostic performance of the test, and confirmed the previously found high sensitivity, specificity, positive and negative predictive values. When evaluating the performance of the test, it is very important to choose a proper population of patients, since the application of screening depends on the individual cancer risk, which is high in genetic syndromes or in those with personal history of colorectal neoplasms, and intermediate in individuals with a positive family history. Approximately 75% of the population are at average risk, and are therefore potential candidates for screening. Therefore, a particular test will have a higher predictive value when applied to a population with a higher prevalence of the disease being studied, as in our case where the test was applied to selected population of patients with already existing colorectal cancer and to patients with previously resected colon tumors or adenomas. Thus, the near perfect performance of the test applied to our small cohort of patients was not a surprise.

In all the studies utilizing the GO-Schiff technology, the test was performed on mucin of the rectum. However, although samples were collected from the rectum, not only rectal cancer, but cancers at distant locations also showed positive results. As a logical explanation is the theory of "field-effect" phenomenon, proposed by Shamsuddin *et al.* [12]. Despite the current dogma that the invasive colon cancer develops only from adenomatous polyp [49], it is also well recognized that carcinogenic factors are important in the pathogenesis of colorectal cancer. Since most cancers in

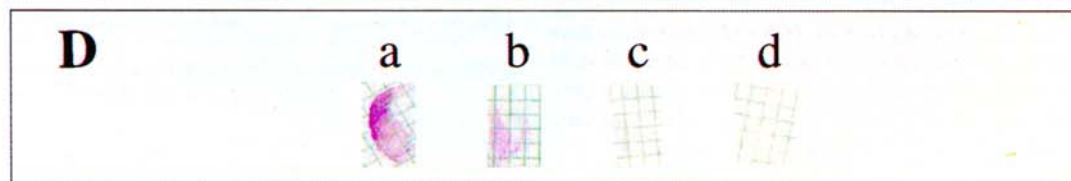
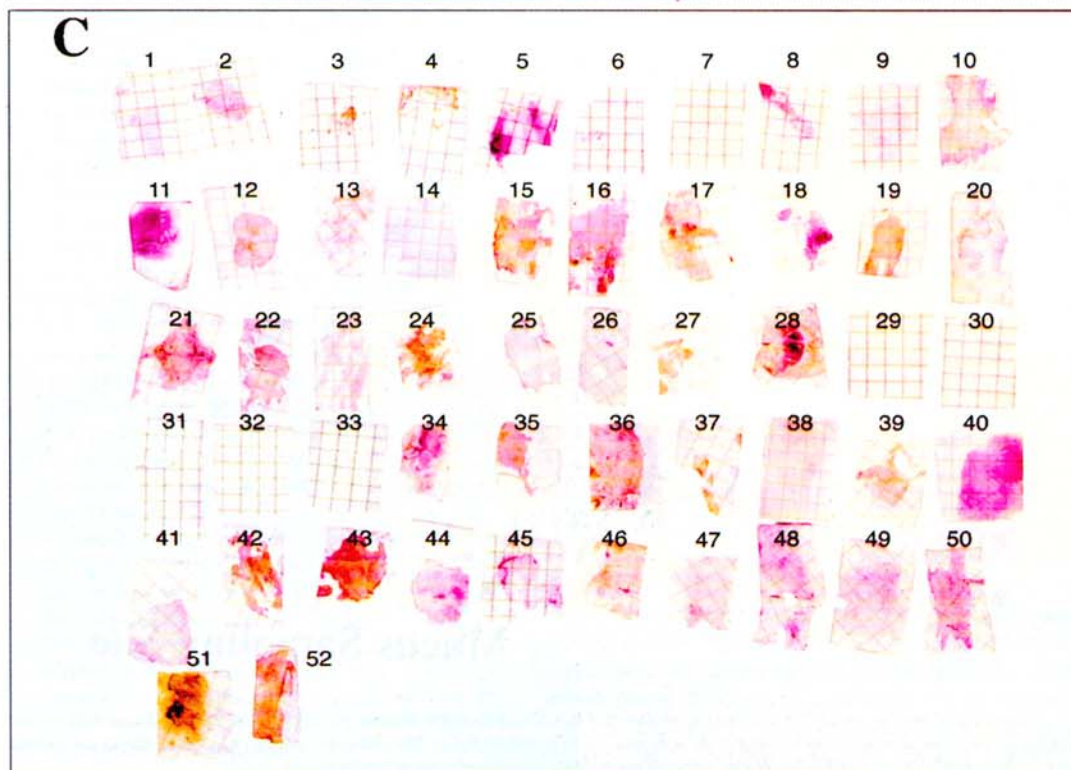
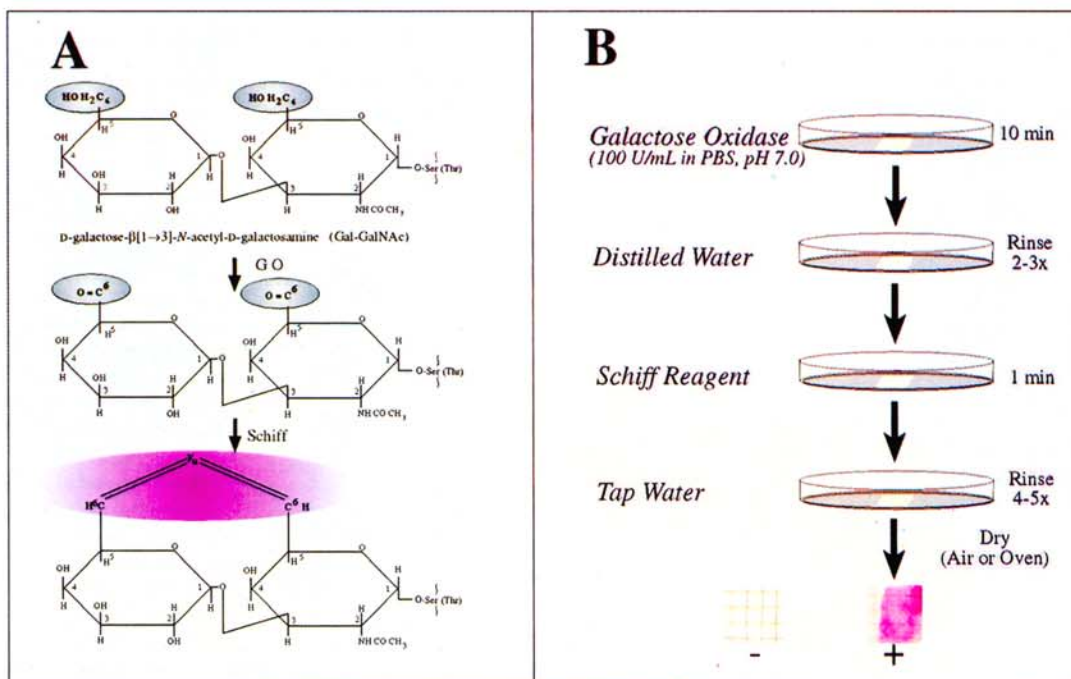
Figure 1. Galactose Oxidase - Schiff Assay: Principle, Protocol and Results.

Panel A. Principle of the GO Test. The enzyme galactose-oxidase specifically oxidizes C-6 hydroxyl groups of D-galactopyranose and N-acetyl-D-galactosamine residues of D-galactose-([1 $\rightarrow$ 3]-N-acetyl-D-galactosamine. This generates two vicinal aldehyde groups that react with basic fuchsin (Schiff's reagent) to give magenta/purple coloration.

Panel B. The Protocol of the Test. Mucus sample smeared onto nitrocellulose membrane reacts with D-galactose oxidase (GO), 100 U/mL in PBS, pH7.0, for 10 min at room temperature. After brief wash with distilled water, the samples are reacted with Schiff reagent (1% basic fuchsin) for 60 sec. Membranes are then rinsed in running tap water, air dried and evaluated for color reaction.

Panel C. The Presence of Gal-GalNAc in a Rectal Mucin. Rectal mucus samples from a randomly selected group of patients with various colorectal malignancies and patients free of colonic disease were smeared on nitrocellulose filters, and developed by GO test, showing the various intensity of magenta color. Diagnoses and results, scored from 0 to 4+, are given in Table II.

Panel D. The Conformation of Mucin on Negative Samples by Periodic Acid. To distinguish whether a negative reaction was due to the absence of sugar Gal-GalNAc, or due to absence of mucin itself (technical negative), negative samples were reacted with periodic acid-Schiff to document the presence of mucin. a) Positive control sample after periodic acid-Schiff reaction indicating the presence of mucin, b) GO negative sample showing presence of mucin after periodic acid-Schiff reaction, c) and d) GO negative samples which do not show presence of any mucin after periodic acid-Schiff reaction (technical negative).



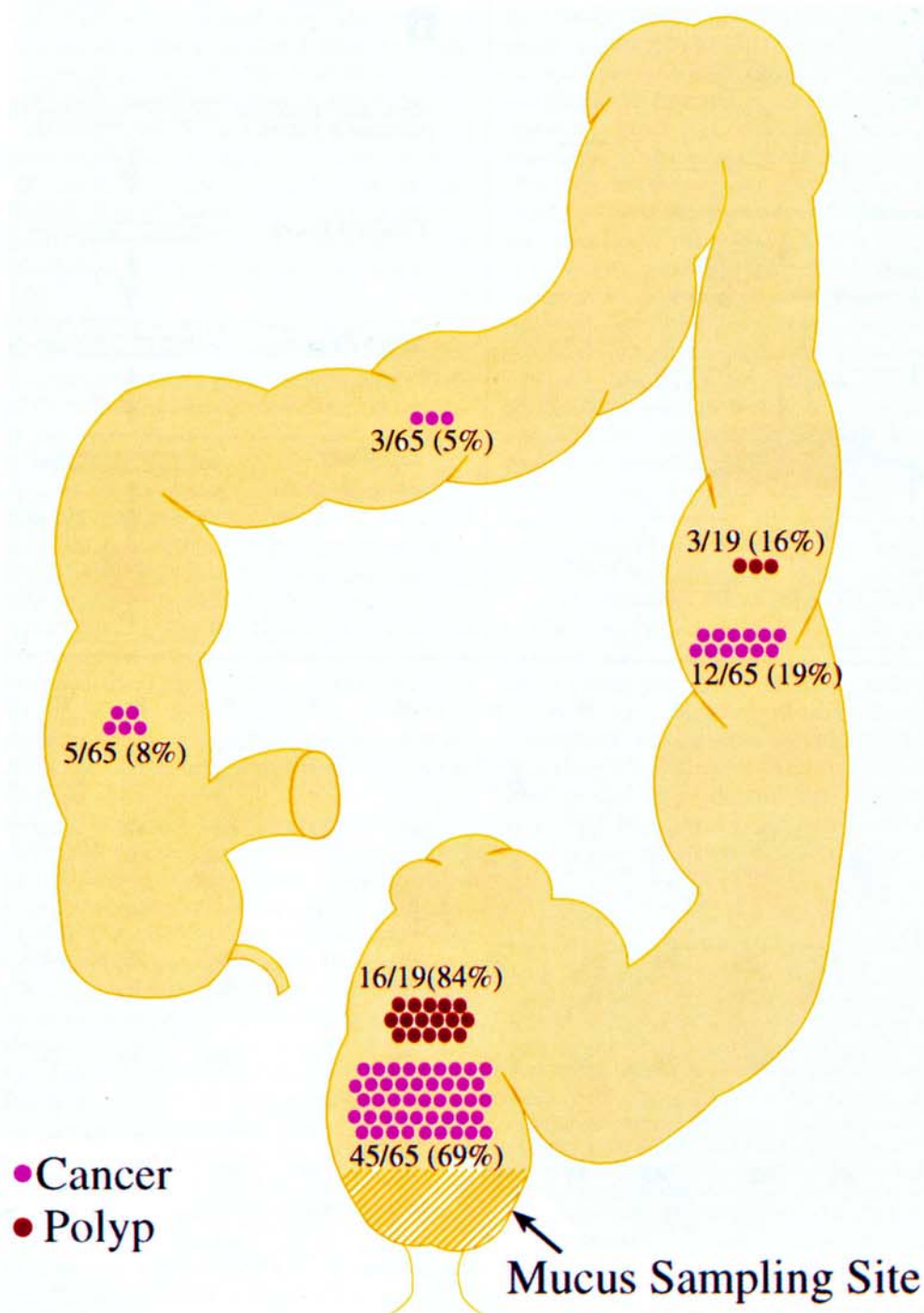


Figure 2. Distribution of the Colorectal Cancers and Polyps Detected by GO-Schiff Assay. Most of the carcinomas (o) were found in the rectum (69%) and the left colon (19%), while only 8% were found in the right and 5% in transverse colon. The location of polyps (•) were similar as carcinomas, located in the rectum (84%) and left colon (16%). The mucus sampling site is shown by hatched mark at the anorectum.

Table II. Diagnoses and GO Reactivity of Rectal Mucus on a Nitrocellulose Filters

Sample#	Clinicopathological Findings	Results of the Test*
1.	Ulcerative colitis	+
2.	Subacute proctitis	++
3.	Sarcoma of the rectum	+
4.	Carcinoma of the rectum	+
5.	Sigmoid polyp	++++
6.	Villous polyp of the rectum (dysplasia)	+
7.	Chronic prostatitis	0
8.	Rectal polyp (dysplasia, grade III)	++
9.	Rectal polyp (dysplasia, grade II)	+
10.	Carcinoma of the rectum	++
11.	Carcinoma of the rectum	++++
12.	Carcinoma of the rectum	++
13.	Sigmoid polyp	+
14.	Carcinoma of the rectum	+
15.	Sigmoid carcinoma	+
16.	Recurrent carcinoma of the rectum	+++
17.	Carcinoma of the rectum	+
18.	Rectal polyp (dysplasia, grade III)	+++
19.	St. post res., adenocarcinoma of the rectum	+
20.	St. post res., carcinoma of the rectum	+
21.	Carcinoma of the rectum	++
22.	Carcinoma of the rectum	++
23.	Carcinoma of the rectum	+
24.	Carcinoma of the rectum (feces cont → repeat)	+
25.	Carcinoma of the rectum, sigmoid polyp	+
26.	Carcinoma of the rectum	+
27.	Carcinoma of the rectum	+
28.	Carcinoma of the rectum, inoperable	++
29.	Chronic prostatitis	0
30.	Chronic prostatitis	0
31.	Chronic prostatitis	0
32.	Chronic prostatitis	0
33.	Chronic prostatitis	0
34.	Adenocarcinoma of the rectosigmoid	+++
35.	Adenocarcinoma of the sigmoid	++
36.	Adenocarcinoma of the sigmoid	++
37.	Adenocarcinoma of the rectum	+
38.	Adenocarcinoma of the rectosigmoid	+
39.	St post res., carcinoma of the rectosigmoid	+
40.	Carcinoma of the descending colon	++++
41.	Sigmoid carcinoma	+
42.	Sigmoid carcinoma	+
43.	Carcinoma of the transverse colon	+
44.	Carcinoma of the rectum	+++
45.	Sigmoid carcinoma	++
46.	Carcinoma of the rectosigmoid	+
47.	St post res., rectal villous polyp	+
48.	Carcinoma of the rectum	+
49.	Carcinoma of the rectosigmoid	+
50.	Carcinoma of the rectosigmoid	++
51.	Carcinoma of the sigmoid repeat (feces contamination)	
52.	Carcinoma of the rectum (feces cont → repeat) +	

* GO Reactivity is shown on Figure 1C. The intensity of Gal-GalNAc expression was scored from 0 - 4+.

Table III. GO Reactivity in a follow-up after resective surgery of colorectal cancer

Patient category	GO Test positive	GO Test negative
Status post resection of	32/53	21/53
Colorectal carcinoma (n=53)	(60%)	(40%)
Tumor recurrence	5/3	20/21
Within a year	(16%)	(0%)

humans are believed to be the result of our exposure to environmental carcinogens which are excreted through the large intestine or urinary bladder, it can be expected that these carcinogens and/or their metabolites cause changes in those organs, too. Indeed these changes were also observed in carcinogen-treated rodents prior to the development of neoplasia, suggesting that these mucin changes are primarily responses to carcinogenic stimuli [10,50]. From the perspective of cellular adaptation, Carter *et al.* have proposed that GO-Schiff reactivity in the morphologically appearing normal colon would be a dosimeter of exposure to an endogenous or exogenous carcinogen [51].

The existence of "field-effect" phenomenon may provide insight into the recurrence of cancer in patients who have undergone total resection of the lesion. GO reactivity in a follow-up study of patients after surgery of colorectal cancer showed evidence of the "field-effect". The positive reactions were obtained in the samples from patients after tumor resection; although malignant tissue was removed, biochemical changes still persisted in the morphologically normal looking tissue. The positive GO test and the presence of the marker indicate that the cells are still abnormal and that these individuals are considered to be at high risk; confirmation of the "field-effect" was evident in a group of patients tested positive with GO, some of whom developed recurrent cancer. Appropriate prophylactic chemopreventive agent in these patients could reduce the risk and even reverse the cancer progression (secondary prevention) [18].

GO reactivity was evident in the cases with ulcerative colitis, a predisposing condition for a colonic cancer. In three separate prospective studies of humans and marmosets with colitis, greater degrees of inflammation were associated with both increased expression of Gal-GalNAc and subsequent development of colorectal cancer [52, 54]. Shamsuddin *et al.* [8] and Carter *et al.* [51] also observed changes in colonic mucin in ulcerative colitis patients. PNA-reactive glycoconjugates in rectal mucus were found in 60% patients with ulcerative colitis [55]. In a follow-up study of a patient with subacute proctitis and with a persistently positive GO test, a sigmoid polyp was found. Thus, some subjects who are currently "false" positive by GO-Schiff test may eventually

show cancer or polyp at a later date. Close follow-up observation and repeated GO-Schiff test are necessary, since some of these "false" positives might turn to true positives.

Evaluating the intensity of the stain in histological sections by semiquantitative analysis using image cytometry, Carter *et al.* [51] have noticed that GO reactivity decreased with neoplastic progression and malignant conversion in the neoplasm and increased in the normal mucosa of patients with progressively more advanced disease. Therefore, they suggest that the GO-Schiff reactivity in the normal colon is a dosimeter of exposure to carcinogens or cytotoxicants, rather than an oncodevelopmental cancer-associated antigen [51]. Using naked eye, we did not find any correlation between intensity of the staining and tumor location, grade or with Duke's stage. In any event, additional studies for quantitative and objective evaluation of stain intensity with better technological assistance will be needed to eliminate errors due to subjective interpretation of test results.

Aside from the simplicity, and high accuracy, there are some additional properties that make this test even more attractive. A remarkable stability of collected samples, which can be kept for several years, without losing reactivity, makes it practical for transportation or shipment. In this study our samples were collected over the period of several months and sustained the transatlantic shipments in regular mail.

Moreover, GO test can have direct application to human chemoprevention programs as a prognostic marker. There are several criteria that biomarkers have to satisfy. Gal-GalNAc detected by GO fits the expected biological mechanisms and satisfies almost all of them, since it is differentially expressed in normal and high-risk tissue, it is closely linked to the causal pathway for the cancer, modulated by chemoprevention agents, may be assayed reliably and easily [56]. Abnormal expressions of mucin have been studied as newer intermediate biomarkers of increased risk for colorectal cancer, that can be modulated by chemopreventive agents [56-58]. When HT-29 human colon carcinoma cells were treated with inositol hexaphosphate, a chemopreventive agent, the expression of Gal-GalNAc was down-regulated *in vitro* [59]. Although our data suggest the potential of clinical utility of this marker detected by GO at high risk population, the impact of chemopreventive agents on its modulation remain to be tested in a population-based study.

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